

Siesta Band Unfolding Document

A manual for the BUFDeepH code.

Quinn Hsu

Last updated: **April 29, 2025**

Contents

Section 1: Band unfolding method	1
Section 2: Unfolding Method1	2
Section 3: Unfolding Method2	2

Here is one band unfolding method to transfer from the deeph-format Hamitonian data to a unfolded band structure.

Here are two methods:

- Method1 is the general method, which is siutable for almost all cases.
- Method2 supports for two-dimensional Moire honeycomb lattice now, which is my original purpose. Actually, for other cases, only minor modifications are needed, you can try it by yourself.

Section 1: Band unfolding method

Hamilton basis:

$$\langle \mathbf{r} | \alpha, \mathbf{k} \rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \phi(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_{\alpha}) \quad [1]$$

Bloch wfc:

$$\begin{aligned} |n, \mathbf{k}\rangle &= \sum_{\alpha} \varphi_{n\alpha}(\mathbf{k}) |\alpha, \mathbf{k}\rangle \\ \langle \mathbf{r} | n, \mathbf{k} \rangle &= \frac{1}{\sqrt{N}} \sum_{\alpha, \mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{R}} \phi(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_{\alpha}) \varphi_{n\alpha}(\mathbf{k}) \end{aligned} \quad [2]$$

Define

$$\phi(\mathbf{p}) = \int d^3\mathbf{r} e^{-i\mathbf{p} \cdot (\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_{\alpha})} \phi(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_{\alpha}) \quad [3]$$

as the Fourier transform of $\phi(\mathbf{r})$, then

$$\begin{aligned} \langle \mathbf{p} | n, \mathbf{k} \rangle &= \frac{1}{N} \sum_{\alpha, \mathbf{R}} \int d^3\mathbf{r} e^{-i\mathbf{p} \cdot \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{R}} \phi(\mathbf{r} - \mathbf{R} - \boldsymbol{\tau}_{\alpha}) \varphi_{n\alpha}(\mathbf{k}) \\ &= \frac{1}{N} \sum_{\alpha, \mathbf{R}} e^{i(\mathbf{k} - \mathbf{p}) \cdot \mathbf{R}} e^{-i\mathbf{p} \cdot \boldsymbol{\tau}_{\alpha}} \phi(\mathbf{p}) \varphi_{n\alpha}(\mathbf{k}) \\ &= \sum_{\alpha, \mathbf{G}} \delta_{\mathbf{p} |, \mathbf{k} + \mathbf{G}} e^{-i\mathbf{p} \cdot \boldsymbol{\tau}_{\alpha}} \phi(\mathbf{p}) \varphi_{n\alpha}(\mathbf{k}) \end{aligned} \quad [4]$$

For ARPES, the intensity is given by

$$\begin{aligned}
I(\mathbf{p}, E) &\propto \sum_{n, \mathbf{k}} |\langle \mathbf{p} | \mathbf{A} \cdot \hat{\mathbf{v}} | n, \mathbf{k} \rangle|^2 \delta(E - \varepsilon_{n\mathbf{k}}) \\
&= \sum_{n, \mathbf{k}} |\mathbf{A} \cdot \sum_{\alpha, \beta, \mathbf{k}'} \langle \mathbf{p} | \alpha, \mathbf{k}' \rangle \langle \alpha, \mathbf{k}' | \hat{\mathbf{v}} | \beta, \mathbf{k}' \rangle \langle \beta, \mathbf{k}' | n, \mathbf{k} \rangle|^2 \delta(E - \varepsilon_{n\mathbf{k}}) \\
&= |\phi(\mathbf{p})|^2 \sum_{n, \mathbf{k}} |\mathbf{A} \cdot \sum_{\alpha, \beta, \mathbf{G}} \delta_{\mathbf{p} |, \mathbf{k} + \mathbf{G}} e^{-i\mathbf{p} \cdot \boldsymbol{\tau}_\alpha} \langle \alpha, \mathbf{k} | \partial_{\mathbf{k}} \hat{H} | \beta, \mathbf{k} \rangle \varphi_{n\beta}(\mathbf{k})|^2 \delta(E - \varepsilon_{n\mathbf{k}}) \\
\text{or} &= |\phi(\mathbf{p})|^2 \sum_{n, \mathbf{k}, \alpha} |(\mathbf{A} \cdot \mathbf{p})^2 \delta_{\mathbf{p} |, \mathbf{k} + \mathbf{G}} e^{-i\mathbf{p} \cdot \boldsymbol{\tau}_\alpha} \varphi_{n\beta}(\mathbf{k})|^2 \delta(E - \varepsilon_{n\mathbf{k}})
\end{aligned} \tag{5}$$

To simplify the problem, we regard the real spherical harmonic basis as the delta function.

For unfolding, similarly, we have

$$\begin{aligned}
I(\mathbf{p}, E) &\propto \sum_g \sum_{N, \mathbf{K}} |\langle \mathbf{p} + \mathbf{g} | N, \mathbf{K} \rangle|^2 \delta(E - \varepsilon_{N\mathbf{K}}) \\
&= |\phi(\mathbf{p})|^2 \sum_{g, N, \mathbf{K}, \alpha, G} |\delta_{\mathbf{p} + \mathbf{g}, \mathbf{K} + \mathbf{G}} e^{-i(\mathbf{p} + \mathbf{g}) \cdot \boldsymbol{\tau}_\alpha} \varphi_{N\alpha}(\mathbf{K})|^2 \delta(E - \varepsilon_{N\mathbf{K}})
\end{aligned} \tag{6}$$

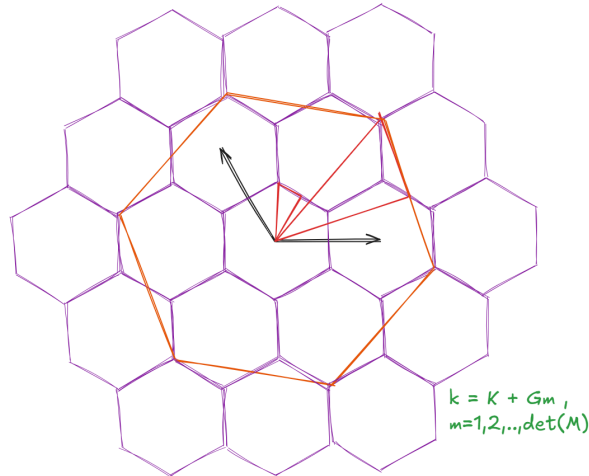
where capital letters for moire cell and small letters for primitive cell.

Section 2: Unfolding Method1

Using eq.6, for a certain $\mathbf{p}$ in space, we can find the corresponding $\mathbf{K}$ and $\mathbf{G}$, ignore the sumover for $\mathbf{G}$, then we can calculate the intensity $I(\mathbf{p})$, with three loops of sumover: 1. $\sum_{N\mathbf{K}}$; 2. $\sum_g$; 3. $\sum_\alpha$.

Section 3: Unfolding Method2

Back to eq.6, in fact, we project the Bloch wave function using a set of plane waves. From the perspective of band folding, the bands at $\det(M)$ G-points in a pBZ are all folded into an mBZ. Conversely, we only need to distinguish which G-point each band comes from, that is, to look at the weight with the largest projection component.



Step1 For each $\mathbf{p}$ in the KPATH, find $\mathbf{p} = \mathbf{K} + \mathbf{G}_0$, where $\mathbf{K}$ in the (0,0)'s mBZ.

Step2 For the $\mathbf{K}$ given in the Step1, find all $\mathbf{k} = \mathbf{K} + \mathbf{G}_m$, where $m = 1, 2, \dots, \det(M)$.

Step3 For each $\mathbf{k}$ given in the Step2, calculate weight w_m and save maximum $\max(w_m)$.

Step4 Compare $\max(w_m)$ with w_0 in Step1, if w_0 is maximum, then accept this unfolding action.